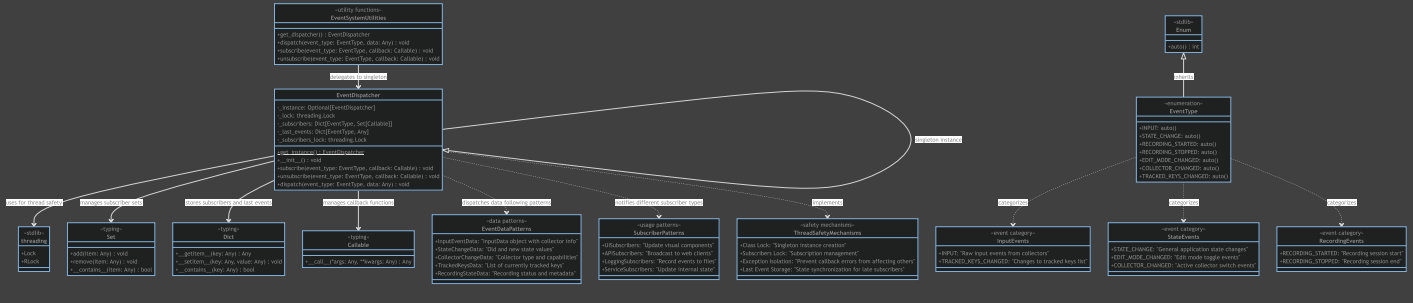

-utility functions-
EventSystemUtilities
+get_dispatcher() : EventDispatcher
+dispatch(event_type: EventType, data: Any) : void
+subscribe(event_type: EventType, callback: Callable) : void
+unsubscribe(event_type: EventType, callback: Callable) : void
delegates to singleton
EventDispatcher
- _instance: Optional[EventDispatcher]
- _lock: threading.Lock
- _subscribers: Dict[EventType, Set[Callable]]
- _last_events: Dict[EventType, Any]
- _subscribers_lock: threading.Lock
+ get_instance() : EventDispatcher
+ __init__() : void
+subscribe(event_type: EventType, callback: Callable) : void
+unsubscribe(event_type: EventType, callback: Callable) : void
+dispatch(event_type: EventType, data: Any) : void
singleton instance
-utils-
Enum
+to(obj) : Int
inherits
-enumeration-
EventType
+INPUT: auto()
+STATE_CHANGE: auto()
+RECORDING_STARTED: auto()
+RECORDING_STOPPED: auto()
+EDIT_MODE_CHANGED: auto()
+COLLECTOR_CHANGED: auto()
+TRACKED_KEYS_CHANGED: auto()
calls for thread safety
-stdlib-
threading
+Lock
+RLock
manages subscriber sets
-typing-
Set
+add(item: Any) : void
+remove(item: Any) : void
+ __contains__(item: Any) : bool
stores subscribers and last events
-typing-
Dict
+ __getitem__(key: Any) : Any
+ __setitem__(key: Any, value: Any) : void
+ __contains__(key: Any) : bool
manages callback functions
-typing-
Callable
+ __call__(*args: Any, **kwargs: Any) : Any
dispatches data following patterns
-data patterns-
EventDataPatterns
+InputEventData: "Specifies objects with collector info"
+StateChangeData: "Old and new state values"
+CollectorChangeData: "Collector type and capabilities"
+TrackedKeysData: "List of currently tracked keys"
+RecordingStateData: "Recording status and metadata"
notifies different subscriber types
-usage patterns-
SubscriberPatterns
+UISubscribers: "Update visual components"
+APISubscribers: "Broadcast to web clients"
+LoggingSubscribers: "Record events to files"
+ServiceSubscribers: "Update internal state"
implements
-safety mechanisms-
ThreadSafetyMechanisms
+Class Lock: "Singleton instance creation"
+Subscribers Lock: "Subscription management"
+Exception Isolation: "Prevent callback errors from affecting others"
+Last Event Storage: "State synchronization for late subscribers"
categorizes
-event category-
InputEvents
+INPUT: "Raw input events from collectors"
+TRACKED_KEYS_CHANGED: "Changes to tracked keys list"
categorizes
-event category-
StateEvents
+STATE_CHANGE: "General application state changes"
+EDIT_MODE_CHANGED: "Edit mode toggle events"
+COLLECTOR_CHANGED: "Active collector switch events"
categorizes
-event category-
RecordingEvents
+RECORDING_STARTED: "Recording session start"
+RECORDING_STOPPED: "Recording session end"